**UNITED STATES DISTRICT COURT**
**DISTRICT OF MASSACHUSETTS**

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| UNITED STATES OF AMERICA | ) | |
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| | ) | |
| v. | ) | Docket No. 1:24-cr-10259-DJC |
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| MATTHEW FARWELL | ) | |

**DEFENDANT'S SUPPLEMENTAL NOTICE OF EXPERT TESTIMONY**

Daniel Loper, an experienced digital forensic expert, submitted a signed Notice of Intent to Offer Expert Testimony in this case on June 1, 2026. On August 20, 2026, additional digital discovery was disclosed to the defense by the government to wit: a 50,643-page Blackbag PDF report of decedent Sandra Birchmore's MacBook Air laptop.

Two letters/notices regarding this disclosure were provided to the defense by the government on August 21st and 24th, 2026 respectively. Based on this additional digital discovery and disclosures regarding the collection of this data by law enforcement, I may offer additional testimony regarding the following:

- On the evening of February 01, 2021, the MacBook Air had a power on/off event at 9:41:43 PM
- Immediately after the power on/off event on February 1, 2021, a message sync'd to the laptop placing an attachment on the device at 9:41:47 PM.
- This sync'd message is the same message that the government references in its "Statement of Facts" as being received by Ms. Birchmore at 9:24 PM on her iPhone 12.
- The February 1, 2021 activity was discovered on the MacBook Air, forensic extraction "Birchmore_Mac.aff4" provided by the government in discovery.
- This MacBook Air extraction was performed on December 06, 2021.

- On August 20, 2026 the government provided defense with a 50,643-page Blackbag PDF report of Sandra Birchmore's MacBook Air.
- A letter provided by the government dated August 21, 2026 states that the Massachusetts State Police (MSP) Trooper Nicholas Guarino of the Norfolk District Attorney's office performed an extraction of the MacBook Air on February 10, 2021.
- On August 31, 2026 defense counsel furnished me with an MSP Division of Investigative Services memo authored by Sergeant John M. Fanning, #3253, dated August 16, 2021 (USAO_008878). According to this memo, Sgt. Fanning and other MSP troopers accessed Sandra Birchmore's laptop on February 9, 2021, before turning the device over to Trooper Guarino.
- The Blackbag PDF report was created on September 29, 2021.
- The Blackbag PDF report does not contain all the data that existed on Ms. Birchmore's laptop.
- The report is a "tagged" artifacts PDF document created from the extraction, and the laptop extraction existed on September 29, 2021.
- The significance of the original extraction is important because Apple devices create files called Unified logs[1] and FSEvents[2].
- Both the Unified logs and FSEvents are volatile data sources that over time get written over by the system and therefore it is extremely important that capture these as close to the time of incident.

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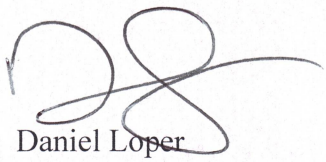
[1] Pete Markham and Will Cornwell, "Unified Logging on Apple Devices: A Guide for Education IT," *Apple Education Community*, October 23, 2025, https://education.apple.com/story/250014802

[2] "Investigating macOS File System Events: The Hidden Forensic Trail," *CyberEngage*, April 2, 2025, https://www.cyberengage.org/post/investigating-macos-file-system-events-the-hidden-forensic-trail

- The government disclosed in a letter dated August 20, 2026, that Stoughton Police Deputy Police Chief Brian Holmes was manually looking at the MacBook Air between November 1 – 14, 2021 during an internal affairs investigation.
- A second letter dated August 24, 2026 further stated that Deputy Police Chief Brian Holmes was actually using the device from October 6, 2021 to December 2021.
- The government's letter states that 35 files had data removed, but only identify 18 of the items in which there are duplicates.
- The government indicated that their expert's Jessica Hyde and others at the FBI evaluated the FSEvents, but no reports of their experts have been disclosed.
- A high-level examination of the forensic extraction of the MacBook Air shows there are forty-nine FSEvents.
- The date of the first FSEvent is September 28, 2021, one day prior to the Blackbag PDF report being created.
- There are twelve FSEvents created December 06, 2021 which most likely are attributed to the forensic extraction performed, but further analysis is required.
- The forensic examination of the "Birchmore_Mac.aff4" extraction shows that the devices Unified Logs and FSEvents from February 01, 2021 have been lost due to device activity by users of law enforcement and therefore potentially relevant data has been lost.
- The August 24, 2026 letter indicates that MSP Lieutenant John Fanning performed a manual review of the MacBook Air prior to the December 6, 2021 extraction being created.
- Manually reviewing digital devices can destroy digital evidence.
- It is industry best practice to perform analysis on forensic extractions unless there are exigent circumstances or an extraction cannot be performed in order to not spoliate evidence.

- Due to the newly disclosed reports and discovery, I am continuing to analyze digital data, and any relevant testimony may change accordingly.
- The MacBook Air PDF is greater than 50,000 pages and it will take considerable effort to compare and contrast its changes with the extraction performed on December 6, 2021.

Signed under the pains and penalties of perjury this the 31st day of August 2026



Daniel Loper